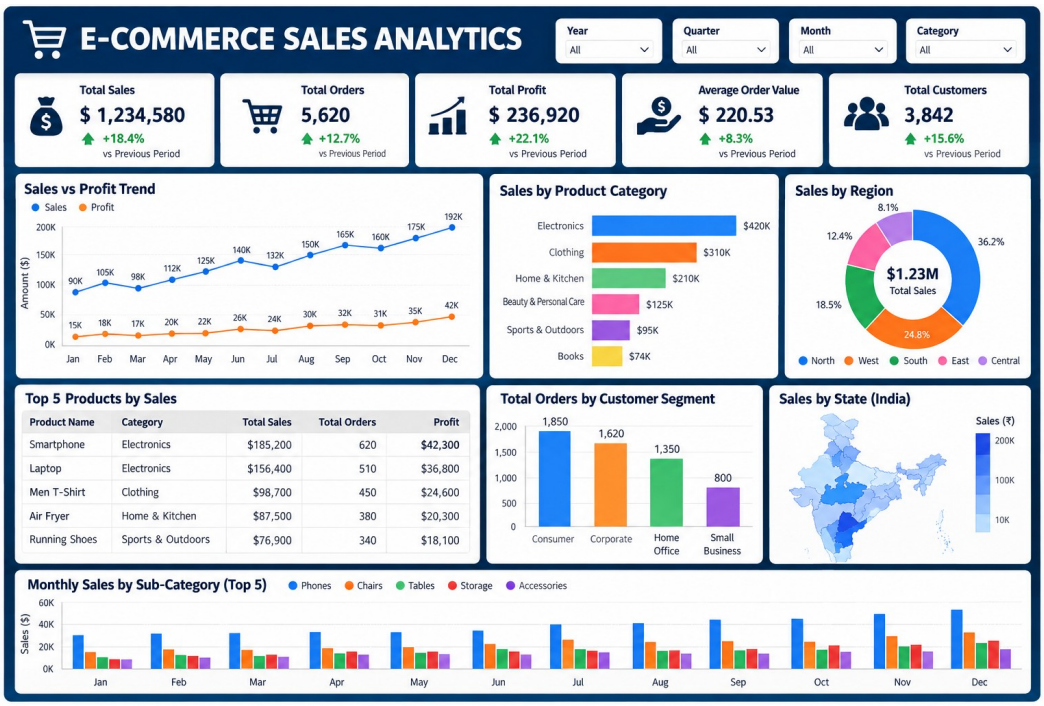
E-Commerce Sales Analytics Dashboard

Visual Explanation Guide | Microsoft Power BI | 5 Pages

1. Dashboard Overview

This dashboard brings together key information about an online store's sales, profit, orders, customers, products, and regions. It is designed to help a business understand what is selling, where sales are coming from, and how performance changes over time.



Important data note: The figures in this image are illustrative sample values. They are not verified company data. The image uses dollar-denominated sales KPIs while the map legend displays rupees, so the currency labels should be standardized before this is used as a real report.

Main objective

To convert transaction data into clear visuals so that a user can monitor performance, compare categories, identify patterns, and make informed decisions.

Questions this dashboard helps answer

- How much sales and profit were generated?
- Which product categories and products contribute most to sales?
- Which regions contribute the largest share of sales?
- Which customer segments place the most orders?
- How do sales vary by month and sub-category?

2. KPI Cards and Pie/Donut Chart

KPI cards provide a quick summary before the user studies the detailed charts. The donut chart then explains how total sales are divided among regions.

KPI Cards

KPI	Sample Value	What it means	Why it matters
Total Sales	$1,234,580	Total revenue from sales.	Tracks overall business scale.
Total Orders	5,620	Number of orders, ideally counted distinctly by Order_ID.	Shows order volume and demand.
Total Profit	$236,920	Total profit in the selected data.	Shows profitability, not just revenue.
Average Order Value	$220.53	Average sales value per order.	Helps track average customer basket size.
Total Customers	3,842	Distinct customers in the selected data.	Shows customer reach.

Pie/Donut Chart: Sales by Region

A pie chart divides a whole into slices. A donut chart does the same but leaves a blank centre, which can be used to display a total. In this dashboard, each colored slice represents a region's share of total sales. The shown shares are approximately North 36.2%, West 24.8%, South 18.5%, East 12.4%, and Central 8.1%. These add to 100%.

How to interpret it

• North has the largest share in the sample visual, so it is the leading region by sales contribution.

• Central has the smallest share. This is a signal to investigate further, not proof that the region is underperforming.

• Use regional sales, profit margin, customer count, and marketing spend together before deciding where to invest.

When to use a pie/donut chart

Use it when you need to show parts of a single whole, ideally with a small number of categories. If there are many categories or values are very similar, a sorted bar chart is often easier to compare.

3. Trend and Category Charts

Line Chart: Sales vs Profit Trend

The line chart compares monthly sales (blue) with monthly profit (orange). Time is placed on the horizontal axis, and the amount is placed on the vertical axis. This makes it easy to see changes across January to December.

• Sales rise overall from about 90K in January to 192K in December in the sample chart.

• Profit also rises overall, from about 15K to 42K, with some month-to-month variation.

• If sales increase while profit stays flat or falls, investigate discounts, costs, returns, and product mix.

Bar Chart: Sales by Product Category

The horizontal bar chart compares sales across categories. Longer bars mean higher sales. The sample image shows Electronics at about $420K, Clothing at $310K, Home & Kitchen at $210K, Beauty & Personal Care at $125K, Sports & Outdoors at $95K, and Books at $74K.

• Electronics leads the sample categories by sales.

• The business can check whether high-sales categories also have healthy profit margins.

• Lower sales do not automatically mean a category should be removed. Consider margin, growth, stock availability, and strategic value.

Why these chart types?

A line chart is effective for trends over time. A bar chart is effective for ranking categories because bar lengths are easy to compare.

4. Product, Customer, Map and Monthly Analysis

Top 5 Products by Sales: Table

The table provides exact values for product name, category, sales, orders, and profit. The sample top products include Smartphone, Laptop, Men T-Shirt, Air Fryer, and Running Shoes.

• Compare both sales and profit: a product with high revenue may have a lower margin.

• Use order count to understand how frequently a product is purchased.

Total Orders by Customer Segment: Column Chart

The column chart compares Consumer (1,850), Corporate (1,620), Home Office (1,350), and Small Business (800) orders in the sample image. Consumer has the highest displayed order count and Small Business the lowest.

Sales by State (India): Map

A map displays sales by geographic location. Stronger shading is intended to indicate higher sales. To build this correctly in Power BI, use a reliable state-level dataset, set the geographic field category to State/Province where appropriate, and ensure that India is identified as the country. Validate ambiguous state names.

Monthly Sales by Sub-Category: Clustered Column Chart

Each month contains multiple colored bars, one for each sub-category. This helps compare sub-category sales month by month and detect seasonal demand. Use the pattern to plan inventory, purchasing, and promotions.

Visual accuracy note

The sample image mixes dollar labels on several visuals with a rupee symbol on the map legend. Standardize the currency and confirm that all charts use the same dataset and filters before presenting the dashboard as a finished analytical report.

5. Filters, DAX Measures and Presentation

Filters / Slicers

Year, Quarter, Month, and Category slicers let users narrow the report to a selected period or product group. When slicers are connected correctly, KPI cards and charts update together. Test combinations of filters and confirm that each visual responds as expected.

Suggested dataset columns

Order_ID, Order_Date, Customer_ID, Product_Name, Category, Sub_Category, Region, State, Customer_Segment, Sales, Profit, and Quantity.

Example DAX measures

Measure	DAX expression
Total Sales	Total Sales = SUM(Orders[Sales])
Total Profit	Total Profit = SUM(Orders[Profit])
Total Orders	Total Orders = DISTINCTCOUNT(Orders[Order_ID])
Total Customers	Total Customers = DISTINCTCOUNT(Orders[Customer_ID])
Average Order Value	Average Order Value = DIVIDE([Total Sales], [Total Orders], 0)

Short presentation script

Good morning, everyone. My project is an E-Commerce Sales Analytics Dashboard developed using Microsoft Power BI. It helps analyze sales, profit, orders, customers, product categories, regional performance, and monthly trends. The KPI cards summarize the business, while the line, bar, donut, table, column, and map visuals provide different perspectives on performance. The slicers allow users to filter the report by year, quarter, month, and category. The purpose is to turn raw transaction data into useful insights for business decisions. The values shown are illustrative sample values and should be replaced with validated data for real reporting. Thank you.

Conclusion

The dashboard combines summary metrics and detailed visuals. To make it reliable, validate totals, standardize currency, configure the map correctly, and test slicers before presenting it.

Possible viva question: Why use a donut chart? Answer: It shows each region's contribution as a part of total sales and makes the relative share easy to understand.